TMA4130/35 K25

Opg 2,3

Problem 2 Interpolation (10 points)
The following table shows the values of a function $f(x)$ at the following points:

x	$f(x)$
1	2
2	8
3	18
4	32

a) Construct an interpolating polynomial $p(x)$ of minimal necessary degree for the given data using the **Lagrange interpolation formula**. Simplify the resulting polynomial as much as possible.

b) Use $p(x)$ to compute an approximate value of $f(2.5)$.

c) Briefly explain why the interpolating polynomial of minimal necessary degree exists and is unique for this data set.

a) Vi har punktene $(1,2),(2,8),(3,18),(4,32)$

Vi skal finne interpolasjonspolynomet $p(x)$

Lagrange

Hvis vi har punkter $(x_0,y_0),\dots,(x_n,y_n)$
så er interpolasjonspolynomet

$$p(x)=\sum_{i=0}^{n} y_i \ell_i(x) \quad \text{der} \quad \ell_i(x)=\prod_{\substack{j=0\\ j\neq i}}^{n} \frac{x-x_j}{x_i-x_j}$$

Navngi punktene

$x_1=1,\ y_1=2$
$x_2=2,\ y_2=8$
$x_3=3,\ y_3=18$
$x_4=4,\ y_4=32$

Sett opp formelen

$$p(x)=2\ell_1(x)+8\ell_2(x)+18\ell_3(x)+32\ell_4(x)$$

Finn $\ell_1(x)$

$$\ell_1(x)=\frac{(x-x_2)}{(x_1-x_2)}\cdot\frac{(x-x_3)}{(x_1-x_3)}\cdot\frac{(x-x_4)}{(x_1-x_4)}$$

$$\ell_1(x)=\frac{(x-2)}{(1-2)}\cdot\frac{(x-3)}{(1-3)}\cdot\frac{(x-4)}{(1-4)}$$

$$\ell_1(x)=\frac{(x-2)(x-3)(x-4)}{-6}$$

Finn $\ell_2(x)$

$$\ell_2(x) = \frac{(x-x_1)}{(x_2-x_1)} \cdot \frac{(x-x_3)}{(x_2-x_3)} \cdot \frac{(x-x_4)}{(x_2-x_4)}$$

$$\ell_2(x) = \frac{(x-1)}{(2-1)} \cdot \frac{(x-3)}{(2-3)} \cdot \frac{(x-4)}{(2-4)}$$

$$\ell_2(x) = \frac{(x-1)(x-3)(x-4)}{2}$$

Finn $\ell_3(x)$

$$\ell_3(x) = \frac{(x-x_1)}{(x_3-x_1)} \cdot \frac{(x-x_2)}{(x_3-x_2)} \cdot \frac{(x-x_4)}{(x_3-x_4)}$$

$$\ell_3(x) = \frac{(x-1)}{(3-1)} \cdot \frac{(x-2)}{(3-2)} \cdot \frac{(x-4)}{(3-4)}$$

$$\ell_3(x) = \frac{(x-1)(x-2)(x-4)}{-2}$$

Finn $\ell_4(x)$

$$\ell_4(x) = \frac{(x-x_1)}{(x_4-x_1)} \cdot \frac{(x-x_2)}{(x_4-x_2)} \cdot \frac{(x-x_3)}{(x_4-x_3)}$$

$$\ell_4(x) = \frac{(x-1)}{(4-1)} \cdot \frac{(x-2)}{(4-2)} \cdot \frac{(x-3)}{(4-3)}$$

$$\ell_1(x) = \frac{(x-1)(x-2)(x-3)}{6}$$

Sett inn i $p(x)$

$$p(x) = 2\ell_1(x) + 8\ell_2(x) + 18\ell_3(x) + 32\ell_4(x)$$

$$p(x) = 2\frac{(x-2)(x-3)(x-4)}{-6} + \frac{8(x-1)(x-3)(x-4)}{2} + \frac{18(x-1)(x-2)(x-4)}{-2} + \frac{32(x-1)(x-2)(x-3)}{6}$$

$$p(x) = -\frac{1}{3}(x-2)(x-3)(x-4) + 4(x-1)(x-3)(x-4) - 9(x-1)(x-2)(x-4) + \frac{16}{3}(x-1)(x-2)(x-3)$$

$$p(x) = -\frac{1}{3}(x^2-5x+6)(x-4) + 4(x^2-4x+3)(x-4) - 9(x^2-3x+2)(x-4) + \frac{16}{3}(x^2-3x+2)(x-3)$$

$$p(x) = -\frac{1}{3}(x^3-5x^2+6x-4x^2+20x-24) + 4(x^3-4x^2+3x-4x^2+16x-12) - 9(x^3-3x^2+2x-4x^2+12x-8)$$
$$+ \frac{16}{3}(x^3-3x^2+2x-3x^2+9x-6)$$

$$p(x) = -\frac{1}{3}(x^3-9x^2+26x-24) + 4(x^3-8x^2+19x-12) - 9(x^3-7x^2+14x-8) + \frac{16}{3}(x^3-6x^2+11x-6)$$

$p(x) = -\frac{x^3}{3} + 3x^2 - \frac{26x}{3} + 8 + 4x^3 - 32x^2 + 76x - 48 - 9x^3 + 63x^2 - 126x + 72 + \frac{16}{3}x^3 - 32x^2 + \frac{176}{3}x - 32$

$p(x) = -\frac{x^3}{3} + 4x^3 - 9x^3 + \frac{16x^3}{3} + 3x^2 - 32x^2 + 63x^2 - 32x^2 - \frac{26}{3}x + 76x - 126x + \frac{176}{3}x + 8 - 48 + 72 - 32$

$\underline{p(x) = 2x^2}$

b) $f(2,5) \approx p(2,5)$

$p(2,5) = 2(2,5)^2$

$p(2,5) = 12,5$

$\underline{f(2,5) \approx 12,5}$

c) Siden x-verdiene er forskjellige finnes det ved interpolasjonsteoremet et unikt polynom av grad høyst 3 som interpolerer punktene.

Problem 3 Numerical Integration (10 points)

a) Consider the approximation formula

$$\int_0^h f(s)ds \approx h\Big(A_1 f(0) + A_2 f(h/3) + A_3 f(h)\Big).$$

Find coefficients $\{A_i\}_{i=1}^3$ such that the formula becomes exact for all polynomials in P_2, i.e., polynomials of degree less than or equal to 2.

b) i) The Simpson's rule provides an approximation for

$$\int_a^b f(s)ds = \frac{\tilde{h}}{3}\big(f(a) + 4f(\tfrac{a+b}{2}) + f(b)\big) - \frac{\tilde{h}^5}{90} f^{(4)}(\tilde{\zeta})$$

where $2\tilde{h} := (b-a)$ and $\tilde{\zeta}$ lies between a and b.

Starting from an equally spaced partition of the interval $[a, b]$

$$a = x_0 < x_1 < x_2 < \cdots < x_{2m-2} < x_{2m-1} < x_{2m} = b,$$

the composite Simpson's rule is defined by applying the Simpson's rule to each subinterval $[x_{2i}, x_{2i+2}]$, and the results are combined to approximate an integral over the whole interval.

Show that the composite Simpson's rule has the following form

$$\int_a^b f(s)ds = \frac{h}{3}\Big(f(a) + f(b) + 4\sum_{i=0}^{m-1} f(x_{2i+1}) + 2\sum_{i=1}^{m-1} f(x_{2i})\Big) - m\frac{h^5}{90} f^{(4)}(\zeta).$$

where $2h := (x_{2i+2} - x_{2i}) = \frac{b-a}{m}$ and ζ lies between a and b.

ii) Use the composite Simpson's rule for $2m = 8$ to compute an approximation for

$$\int_1^2 \ln(s)ds = 2\ln(2) - 1.$$

iii) Show that the approximation error cannot be larger than 0.000008.

a) Vi har formelen $\int_0^h f(s)ds \approx h(A_1 f(0) + A_2 f(\frac{h}{3}) + A_3 f(h))$

Vi skal finne A_1, A_2, A_3 slik at formelen er eksakt for alle polynomer av grad ≤ 2

Eksakthet for polynomer
At formelen er eksakt for alle polynomer av grad ≤ 2, betyr det at den må stemme for basispolynomene $1, s, s^2$

Linearitet
Hvis formelen stemmer for $1, s, s^2$, så stemmer den for alle polynomer $p(s) = a + bs + cs^2$

Bruk $f(s) = 1$

VS:
$$\int_0^h 1\,ds = h$$

HS:
$$h(A_1 \cdot 1 + A_2 \cdot 1 + A_3 \cdot 1) = h(A_1 + A_2 + A_3)$$

Sett sammen:
$$h = h(A_1 + A_2 + A_3) \Rightarrow A_1 + A_2 + A_3 = 1$$

Bruk $f(s) = s$

VS:
$$\int_0^h s\,ds = \left[\frac{s^2}{2}\right]_0^h = \frac{h^2}{2}$$

HS:
$$h\left(A_1 f(0) + A_2 f\left(\frac{h}{3}\right) + A_3 f(h)\right) = h\left(A_1 \cdot 0 + A_2 \cdot \frac{h}{3} + A_3 h\right) = h^2\left(\frac{A_2}{3} + A_3\right)$$

Sett sammen:
$$\frac{h^2}{2} = h^2\left(\frac{A_2}{3} + A_3\right) \Rightarrow \frac{1}{2} = \frac{A_2}{3} + A_3$$

Sett $f(s) = s^2$

VS:
$$\int_0^h s^2\,ds = \left[\frac{s^3}{3}\right]_0^h = \frac{h^3}{3}$$

HS:
$$h\left(A_1 \cdot 0 + A_2 \cdot \left(\frac{h}{3}\right)^2 + A_3 \cdot h^2\right) = h\left(A_2 \frac{h^2}{9} + A_3 h^2\right) = h^3\left(\frac{A_2}{9} + A_3\right)$$

Sett sammen:
$$\frac{h^3}{3} = h^3\left(\frac{A_2}{9} + A_3\right) \Rightarrow \frac{1}{3} = \frac{A_2}{9} + A_3$$

Ligningssystem

$$\begin{aligned} A_1 + A_2 + A_3 &= 1 \\ \frac{1}{2} &= \frac{A_2}{3} + A_3 \\ \frac{1}{3} &= \frac{A_2}{9} + A_3 \end{aligned} \Rightarrow \begin{aligned} A_1 + A_2 + A_3 &= 1 \\ 0 + \frac{A_2}{3} + A_3 &= \frac{1}{2} \\ 0 + \frac{A_2}{9} + A_3 &= \frac{1}{3} \end{aligned} \Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1/3 & 1 & 1/2 \\ 0 & 1/9 & 1 & 1/3 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1/3 & 1 & 1/2 \\ 0 & 1/9 & 1 & 1/3 \end{array}\right] \underset{\sim}{3R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 3/2 \\ 0 & 1/9 & 1 & 1/3 \end{array}\right] \underset{\sim}{R_3 - \frac{1}{9}R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 3/2 \\ 0 & 0 & 2/3 & 1/6 \end{array}\right]$$

$$\underset{\sim}{\frac{3}{2}R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 3 & 3/2 \\ 0 & 0 & 1 & 1/4 \end{array}\right] \underset{\sim}{\substack{R_1 - R_3 \\ R_2 - 3R_3}} \left[\begin{array}{ccc|c} 1 & 1 & 0 & 3/4 \\ 0 & 1 & 0 & 3/4 \\ 0 & 0 & 1 & 1/4 \end{array}\right] \underset{\sim}{R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3/4 \\ 0 & 0 & 1 & 1/4 \end{array}\right]$$

$$A_1 = 0,\ A_2 = \frac{3}{4},\ A_3 = \frac{1}{4}$$

b) (i) Vi skal vise formelen

$$\int_a^b f(s)\,ds = \frac{h}{3}\left(f(a) + f(b) + 4\sum_{i=0}^{m-1} f(x_{2i+1}) + 2\sum_{i=1}^{m-1} f(x_{2i})\right) - m\frac{h^5}{90} f^{(4)}(\xi)$$

Simpsons metode på ett dobbeltintervall

For intervallet $[x_{2i}, x_{2i+2}]$:

$$\int_{x_{2i}}^{x_{2i+2}} f(s)\,ds = \frac{h}{3}\left(f(x_{2i}) + 4f(x_{2i+1}) + f(x_{2i+2})\right) - \frac{h^5}{90} f^{(4)}(\xi_i)$$

Her er h avstanden mellom to nabopunkter: $x_{j+1} - x_j = h$

Sammensatt Simpson

Vi deler hele intervallet $[a,b]$ i $2m$ delintervaller.

Da bruker vi Simpson på par av intervaller.

Del integralet opp

$$\int_a^b f(s)\,ds = \sum_{i=0}^{m-1} \int_{x_{2i}}^{x_{2i+2}} f(s)\,ds$$

Bruk Simpson på hvert delintervall

$$\int_{x_{2i}}^{x_{2i+2}} f(s)\,ds = \frac{h}{3}\left(f(x_{2i}) + 4f(x_{2i+1}) + f(x_{2i+2})\right) - \frac{h^5}{90} f^{(4)}(\xi_i)$$

Setter dette inn i summen:

$$\int_a^b f(s)\,ds = \sum_{i=0}^{m-1}\left[\frac{h}{3} f(x_{2i}) + 4f(x_{2i+1}) + f(x_{2i+2})) - \frac{h^5}{90} f^{(4)}(\xi_i)\right]$$

Del opp summen

$$\int_a^b f(s)\,ds = \frac{h}{3}\sum_{i=0}^{m-1}\left(f(x_{2i}) + 4f(x_{2i+1}) + f(x_{2i+2})\right) - \sum_{i=0}^{m-1}\frac{h^5}{90}f^{(4)}(\xi_i)$$

Se på funksjonsverdiene

For $i=0$, intervallet $[x_0, x_2]$:

$f(x_0) + 4f(x_1) + f(x_2)$

For $i=1$, intervallet $[x_2, x_4]$:

$f(x_2) + 4f(x_3) + f(x_4)$

For $i=2$, intervallet $[x_4, x_6]$:

$f(x_4) + 4f(x_5) + f(x_6)$

Skriv summen kompakt

$$f(a) + f(b) + 4\sum_{i=0}^{m-1} f(x_{2i+1}) + 2\sum_{i=1}^{m-1} f(x_{2i})$$

Så:

$$\int_a^b f(s)\,ds = \frac{h}{3}\left(f(a) + f(b) + 4\sum_{i=0}^{m-1} f(x_{2i+1}) + 2\sum_{i=1}^{m-1} f(x_{2i})\right) - \sum_{i=0}^{m-1}\frac{h^5}{90}f^{(4)}(\xi_i)$$

Feilleddet

Siden vi har m Simpson-blokker, får vi m feilledd

$$-\sum_{i=0}^{m-1}\frac{h^5}{90}f^{(4)}(\xi_i) = -m\frac{h^5}{90}f^{(4)}(\xi) \quad \text{for en } \xi \in (a,b)$$

Konklusjon

$$\int_a^b f(s)\,ds = \frac{h}{3}\left(f(a) + f(b) + 4\sum_{i=0}^{m-1} f(x_{2i+1}) + 2\sum_{i=1}^{m-1} f(x_{2i})\right) - m\frac{h^5}{90}f^{(4)}(\xi)$$

b) (ii) Vi skal bruke sammensatt Simpson metode med $2m = 8$ til å approksimere $\int_1^2 \ln(s)\,ds$

Sammensatt Simpson metode

Når intervallet deles i $2m$ like delintervaller:

$$\int_a^b f(s)\,ds \approx \frac{h}{3}\left(f(x_0) + f(x_{2m}) + 4\sum_{i=0}^{m-1} f(x_{2i+1}) + 2\sum_{i=1}^{m-1} f(x_{2i})\right)$$

$$h = \frac{b-a}{2m}$$

Finn $a, b, 2m, h$

$a = 1$, $b = 2$, $2m = 8$

$$h = \frac{2-1}{8} = \frac{1}{8}$$

Lag punktene

$x_i = a + ih \qquad x_i = 1 + \frac{i}{8}$

$x_0 = 1$, $x_1 = \frac{9}{8}$, $x_2 = \frac{5}{4}$, $x_3 = \frac{11}{8}$, $x_4 = \frac{3}{2}$, $x_5 = \frac{13}{8}$, $x_6 = \frac{7}{4}$, $x_7 = \frac{15}{8}$, $x_8 = 2$

Sett inn i Simpsons formel

$$\int_1^2 \ln(s)\,ds \approx \frac{h}{3}\left[f(x_0) + f(x_8) + 4\left(f(x_1) + f(x_3) + f(x_5) + f(x_7)\right) + 2\left(f(x_2) + f(x_4) + f(x_6)\right)\right]$$

Siden $f(s) = \ln(s)$

$$\approx \frac{1}{24}\left[\ln(1) + \ln(2) + 4\left(\ln\frac{9}{8} + \ln\frac{11}{8} + \ln\frac{13}{8} + \ln\frac{15}{8}\right) + 2\left(\ln\frac{5}{4} + \ln\frac{3}{2} + \ln\frac{7}{4}\right)\right]$$

$$\underline{\underline{\approx 0{,}386292}} \qquad (2\ln(2) - 1 \approx 0{,}386294)$$

b) (iii) Vi skal vise at feilen i Simpson-approksimasjonen ikke kan være større enn $0{,}000008$ for $\int_1^2 \ln(s)\,ds$ med $2m = 8$

Feilestimat for sammensatt Simpsons metode

$$|E| \le \frac{b-a}{180} h^4 \max_{a \le s \le b} |f^{(4)}(s)| \quad , h = \frac{b-a}{2m}$$

Finn $f^{(4)}(s)$

$f(s) = \ln(s)$

$f'(s) = \frac{1}{s}$, $f''(s) = -\frac{1}{s^2}$, $f^{(3)}(s) = \frac{2}{s^3}$, $f^{(4)}(s) = -\frac{6}{s^4}$

Finn max på [1,2]

$|f^{(4)}(s)| = \frac{6}{s^4}$

Denne er størst når s er minst

$\Rightarrow s = 1$

$\Rightarrow \max_{1 \le s \le 2} |f^{(4)}(s)| = \frac{6}{1^4} = 6$

Sett inn i feilformelen

$|E| \le \frac{b-a}{180} h^4 \max |f^{(4)}(s)|$

$|E| \le \frac{1}{180} \left(\frac{1}{8}\right)^4 \cdot 6$

$|E| \le 8{,}14 \cdot 10^{-6} \approx 0{,}000008$